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DTCS–404

DIPLOMA (CSE)
(FOURTH SEMESTER)
MID SEMESTER
EXAMINATION, March, 2024

DATA STRUCTURES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is data structure ? Categorize the different types of data structures and explain any *one* operation on array with suitable example. (CO1)

OR

(b) What is algorithm explain with an example ? Illustrate the steps involved in designing of any algorithm. (CO1)

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2. (a) Write a C program to search a particular element 'item' from the list of n number using linear search technique. (CO2)

OR

(b) Sort the following sequence of numbers using bubble sort technique and show each step clearly : (CO2)

25, 10, 15, 30, 7, 8, 12, 23

3. (a) Compute the Big Oh of the following function : (CO1)

(i) $f(n) = 3n + 5$

(ii) $f(n) = n^2 - 3n + 5$

OR

(b) What is abstract data type ? Explain with suitable example. (CO1)

4. (a) Write the Algorithm for selection sort and explain the technique with suitable example. (CO2)

OR

(b) Explain the binary search technique using your own data set. Also explain why it is better than linear search technique for sorted data set. (CO2)

5. (a) Suppose m number of different elements of integer data type is stored in an array of size n ($n > m$). If you want to insert an element "50" at array index j ($j < m$), write the steps involved to perform the above operation in the array. (CO2)

OR

(b) Sort the following sequence of numbers using selection sort technique and show each step clearly : (CO2)

10, 5, 7, 9, 13, 18, 3, 4

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